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Full Length Article

Wire and arc additive manufacturing of 308L stainless steel components: Optimization of processing parameters and material properties

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ABSTRACT

In the current study, wire and arc additive manufacturing (WAAM) of thin-walled 308L stainless steel components was reported. Firstly, the influence of the welding current, the voltage, and the travel speed in the WAAM process on the geometry of single weld beads was investigated. These parameters were also optimized for the deposition of 308L steel walls. Secondly, the microstructure and the mechanical characteristic of WAAM 308L steel walls have been explored. The obtained results reveal the optimized that parameters giving the desirable geometry of single weld beads for building 308L walls by the WAAM process. The microstructure of WAAM 308L steel walls mainly features the dendrites of austenite growing vertically and residual ferrite existing in grain boundaries of the austenite matrix. The microhardness of the built material is about 163 $HV_{0.1}$. The UTS, YS and elongation of WAAM 308L walls are 532–553 MPa, 344–353 MPa, and 40–54%, respectively, which are relatively similar to those of wrought 308L stainless steel (UTS: 530–650 MPa, YS: 360–480 MPa, and elongation: 35–45%). Thus, the mechanical properties of WAAM 308L steel walls are considered to be adequate for industrial applications.
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1. Introduction

Additive manufacturing (AM) technology, which uses the electric arc to melt metallic wire, also called wire arc additive manufacturing (WAAM), is widely employed for producing large dimensional metallic components. The WAAM process exhibits significantly high deposition rate, about 3–8 kg/h when compared with 0.1–0.6 kg/h of the powder bead or feed AM systems [1]. The use of metal wire is safer for the operators' health and the environment than the metal powder. Furthermore, the production of metal wire is cheaper than that of metal powder, leading to the reduction in total production costs [2]. In WAAM processes, the arc can be GMAW (gas metal arc welding), PAW (plasma arc welding), and GTAW (gas tungsten arc welding) [3]. Among these technologies, the GMAWAM presents the highest deposition rate, about two or three times higher than those of GTAWAM and PAWAM. Thus, it is more suited to manufacture metal components with large dimensions [4].

Recently, most of the published research focused on exploring the external and internal qualities of components produced by WAAM from titanium, aluminum, and nickel-based alloys [5–8]. For steels, previous studies mostly used 304, 304L and 316L steel (that are austenite stainless steels) and low-carbon steels as the feedstock material. Rafieazad et al. [9] explored the microstructural and mechanical characteristics of multi-bead multi-layer low-carbon steel walls manufactured by the GMAWAM process, while Le [10] and Lu et al. [11] used the GMAWAM systems to build single-bead multi-layer thin-walled low-carbon steel components. These authors observed that the microstructure of GMAWAM low-carbon steel walls changed from layer to layer along the building direction. The microstructure was commonly characterized by granular structures composed of granular ferrite, residual austenite, and a little perlite in grain boundaries. Haden et al. [12] carried out a study in which a GMAWAM system was used to build thin walls from both low-carbon steel (ER70S) and 304 steel. They demonstrated that the tensile strength of both GMAWAM mild steel and 304 steel falls within the same value range as that of wrought low-carbon and 304 steels, respectively. Chen et al. [13] explored the microstructures of GMAWAM 316L walls. The authors observed that the microstructures of GMAWAM 316L walls consist of austenite dendrites and residual ferrite and sigma phases, which exist within the austenite matrix. The austenitic dendrites form large columnar grains growing along the building direction. Wang et al. [14] also clarified the correlation

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between the arc modes, the microstructures and the mechanical properties in the WAAM process of 316L steel. These authors highlighted that the arc mode having lower heat input produced finer microstructures in both the bottom and the upper layers. Wu et al. [4] compared the microstructure and tensile properties of GMAWAM 316L walls deposited by two welding current processes. They found that higher welding currents lead to higher heat input produce coarser structures and influence the hardness of built materials. Gao et al. [15] investigated the variation of microstructure of 9Cr ferrite/martensitic steel walls built by a WAAM process, in which the CMT welding was used. The authors showed that the microstructure of the wall changed from region by region. The bottom zone featured smaller grain sizes than in the middle and the top zone of the wall. The anisotropy in terms of microstructures was one of main factor that caused the difference of hardness and tensile strengths in the building (vertical) direction and the horizontal direction.

To the best authors' knowledge, very limited studies observed the microstructural and mechanical characteristics of WAAM 308L components. 308L steel is an austenite stainless steel, which features a lower level of carbon, high mechanical properties, and high corrosion resistance. Its applications can be found in different sectors, for example gas, oil, manufacturing and mining, steamship building and automobile industries. Certain previous studies used 308LSi and 308L steel as the feedstock material in GTAWAM, laminar plasma AM, and laser metal deposition. For example, Yilmaz and Ugla [16] reported the effect of processing parameters (including the travel speed and the wire feed speed) on the microstructures of GTAWAM 308LSi walls. They observed that the travel speed significantly influenced the texture of residual ferrite and the size of grains, and the wire feed speed had a notable effect on the refinement of grains. This research gives a good insight into the microstructure evolution of GTAWAM 308LSi walls. However, the hardness and the tensile strengths of GTAWAM 308LSi were not investigated. Abioye et al. [17] used the laser metal wire deposition process to build single-track multi-layer 308LSi walls. The authors found that the microstructures of 308LSi walls were characterized by columnar dendrites, which grow vertically along the building direction. More recently, Li et al. [18] employed the laminar plasma and metal wire-based deposition process to manufacture a hollow component with four straight walls. Subsequently, the evolution of microstructures and tensile strengths of the walls were investigated. These authors also observed that the bottom and the middle of the walls mainly consisted of columnar dendrites, whereas the equiaxed structures appeared in the top region. Recently, Laghi et al. [19,20] also employed ER308LSi stainless steel as the wire feedstock in WAAM. Their project focused on the use of WAAM for building 308LSi stainless steel structural components in the civil engineering sector. Le et al. [21] investigated the effect of different cooling methods on the shape and material properties of stainless steel 308L walls manufactured by WAAM.

Overall, the surface appearance and the metallurgical characteristics of GMAWAM 308L walls were rarely reported in the literature. Additionally, previously published studies generally used a set of process parameters that were selected from a value range suggested by the wire manufacturers and/or after a number of runs. Thus, the optimization of processing parameters, which would give desirable geometry and quality of weld beads for the build of thin-walled components, should also be taken into consideration. Taken together, this study focuses on exploring the process stability, the microstructure, and the mechanical characteristics of GMAWAM 308L walls. In the first stage, the influence of main processing parameters including the welding current, the travel speed, and the voltage on the shape and quality of single weld beads was studied by using the response surface methodology (RSM) and ANOVA. A set of optimized process parameters were also determined for the build of 308L steel walls. In the second stage, the microstructure, microhardness and tensile properties of GMAWAM 308L steel walls built with the optimized process parameters were explored. The results obtained in this stage are considered as important evidence to promote the use of GMAWAM 308L stainless steel parts for industrial applications.

2. Experimental procedures

A commercial wire of 308L steel with 1 mm diameter and a number of SS400 steel plates (150 mm × 250 mm × 10 mm in width, in length and in thickness, respectively) were used in the GMAWAM process. The chemical elements of the wire and the substrate were given in Table 1. The welding heat source in the GMAWAM process is a power source of Panasonic (YD-350GR3). For this GMAWAM system, the wire feed speed is automatically adjusted according to the welding current. The wire feed speed is increased with an increase in the welding current. The motion of the welding torch along the deposition paths is performed by a Panasonic TA1400 6-axis robot. During the welding process, a 99.99% argon gas was applied for the shielding. The flow rate of shielding gas is 15 L/min. After a deposition was finished, the specimen was cooled down by the natural air to room temperature.

As the geometry and quality of weld beads are significantly important for building thin-walled components by the WAAM process [22], the first stage aims to explore the effects of main processing parameters, including the welding current, the travel speed, and the voltage on the geometry of weld beads, and determine a set of optimal process parameters that would give the desired quality and shape of single weld bead. In this study, the bead height and the bead width were predicted as functions of processing parameters from the experimental data. For this purpose, the response surface methodology (RSM) and the analysis of variance (ANOVA) [23] were used. The experimental plan was designed by using the Box-Behnken method. Three levels for each factor were selected as presented in Table 2. The working limits of each parameter were selected based on the recommendation of the wire manufacturer. These values were also verified by a number of trial runs to ensure the weldability and avoid the interruption of experiments.

Table 3 shows seventeen trial runs of single weld beads. The length of each welding bead is around 120 mm. The bead height (*BH* in mm) and the bead width (*BW* in mm) of a welding bead were measured at five positions in the steady region of the welding bead (Fig. 1) by using a digital caliper. The average values of *BW* and *BH* were then taken, as reported in Table 3.

In the second stage, to observe the process stability, the microstructures and the mechanical characteristics of GMAWAM 308L walls, two single-bead multi-layer walls have been built with the optimized processing parameters that were obtained from the first stage. The walls were deposited according to the alternating direction strategy with a dwell time of 40 s between adjacent layers [26]. The dwell time was applied to cool down the parts and to transfer the accumulated heat to the environment. The distance between the contact tip of the welding torch and the workpiece is about 12 mm.

Fig. 2 shows two GMAWAM 308L walls built with optimized processing parameters. The first wall (Fig. 2a) with dimensions of around 115 mm × 102 mm in length and in height, respectively, was used to prepare three microstructure test specimens (MST, MSM and MSB), a chemical composition test specimen (CS), and three vertical tensile test specimens (vTS-1, vTS-2 and vTS-3) in the building direction. The second wall with dimensions of 115 mm × 59 mm (Fig. 2b) was used to prepare three horizontal tensile specimens (hTS-1, hTS-2 and hTS-2) in the depositing

Table 1
Chemical composition of the 308L steel wire and low-carbon steel (SS400) substrate.

Element	C	P	S	Cr	Ni	Mo	Cu	Si	Mn	Al	Fe
Metal wire (308L) [24]	0.03 max	0.03 max	0.03 max	19.5–21	9.0–11.0	0.50 max	0.75 max	0.30–0.65	1.0–2.5	–	Balance
Substrate (SS400) [25]	0.05	0.013	0.002	–	–	–	–	0.037	0.46	0.044	Balance

Table 2
Three levels of processing parameters.

Parameter	Levels		
Welding current, I (A)	100	120	140
Voltage, U (V)	17	20	23
Travel speed, v (mm/min)	300	400	500

direction. It is noticed that two side surfaces of the walls were first removed, and then all specimens were extracted by using a wire-cut EDM machine.

The chemical composition of GMAWAM 308L steel walls was analyzed, using a Thermo Scientific Spectrometer (ARL3460 OES). Before the test, this specimen CS was polished and cleaned. Three tests were carried out and the final result (in percentage) is the average value for each element, as presented in Table 4. It is noticed that the main elements of GMAWAM 308L walls have the percentage falling within the same range as that of the 308L wire of the manufacturer (Table 1).

To observe the microstructure, the surfaces of three specimens (MST, MSM and MSB) that are corresponding to the cross-section (P) of the wall were mounted and ground with a series of silicon carbide papers. Subsequently, they were polished with diamond polishing agent of 2.5 μm particles, and chemically etched by using a solution of (10 mL HNO_3, 20 mL HCl, and 30 mL water) in 10 s. The microstructure of GMAWAM 308L walls was explored by the optical microscope (AXIO A2M). The microhardness of the GMAWAM 308L wall was measured by a Vickers microhardness tester (Duramin 2 of Struers) according to the Standard ASTM E384. For each hardness test, a load of 980.7 mN and a dwell time of 10 s were applied. For each specimen, the microhardness was measured at five locations that are distributed on the center line of the cross section. The microhardness at a location is the average value of three different indentations. The spacing between two adjacent indentations is 1 mm.

The tensile testing machine (FU/R DVT-DEVOTRANS) was used to carry out the tensile tests at room temperature. The speed of the cross-head is 20 mm/min. The facture surface morphologies of the tensile specimens were observed by the SEM machine (SU3500).

3. Results and discussion

3.1. Developing the predictive models

In this study, the second order regression equation was chosen to develop the predictive models of bead width and the bead height, Eq. (1):

$$y = c_0 + \sum_{i=1}^{3} c_i x_i + \sum_{i=1}^{3} c_{ii} x_i^2 + \sum_{i,j=1, i\neq j}^{3} c_{ij} x_i x_j \quad (1)$$

where y is the responses – i.e., BW (mm) and BH (mm); x_i and x_j are the input variables, namely the welding current I (A), the voltage U (V), and the travel speed v (mm/min)); c_0, c_i, c_{ii}, and c_{ij} are the coefficients of the developed models. In the current work, the coefficients were estimated by using the Minitab 19 software. The full predictive models of the bead width and the bead height are described in Eqs. (2) and (3), respectively:

$$BW\ (\text{mm}) = -16.190 + 0.143 \times I + 1.027 \times U + 0.508 \times 10^{-3} \times v + 0.375 \times 10^{-3} \times I \times U - 0.016 \times 10^{-3} \times I \times v - 0.263 \times 10^{-3} \times U \times -0.567 \times 10^{-3} \times I^2 - 0.018 \times U^2 + 3.67 \times 10^{-6} \times v^2 \quad (2)$$

$$BH\ (\text{mm}) = 7.575 + 0.085 \times I - 0.623 \times U - 0.014 \times v + 8.33 \times 10^{-6} \times I \times U + 0.024 \times 10^{-3} \times I \times v + 0.048 \times 10^{-3} \times U \times v - 0.292 \times 10^{-3} \times I^2 + 0.013 \times U^2 + 5.223 \times 10^{-6} \times v^2 \quad (3)$$

The adequacy of the predictive models was evaluated by using the ANOVA. The ANOVA was carried out with 5% significance level and 95% confidence level. Table 5 shows the ANOVA results for the

Table 3
Experimental design and measured data.

Run	Input variables			Response	
	I (A)	U (V)	v (mm/min)	BW (mm)	BH (mm)
1	100	17	400	3.79	3.01
2	140	17	400	4.03	3.98
3	100	23	400	5.08	2.49
4	140	23	400	5.41	3.46
5	100	20	300	4.96	3.29
6	140	20	300	5.42	4.22
7	100	20	500	4.19	2.02
8	140	20	500	4.53	3.14
9	120	17	300	4.31	4.21
10	120	23	300	6.02	3.66
11	120	17	500	3.81	3.12
12	120	23	500	5.21	2.62
13	120	20	400	4.97	3.17
14	120	20	400	5.00	3.28
15	120	20	400	4.97	3.30
16	120	20	400	4.87	3.22
17	120	20	400	5.01	3.20

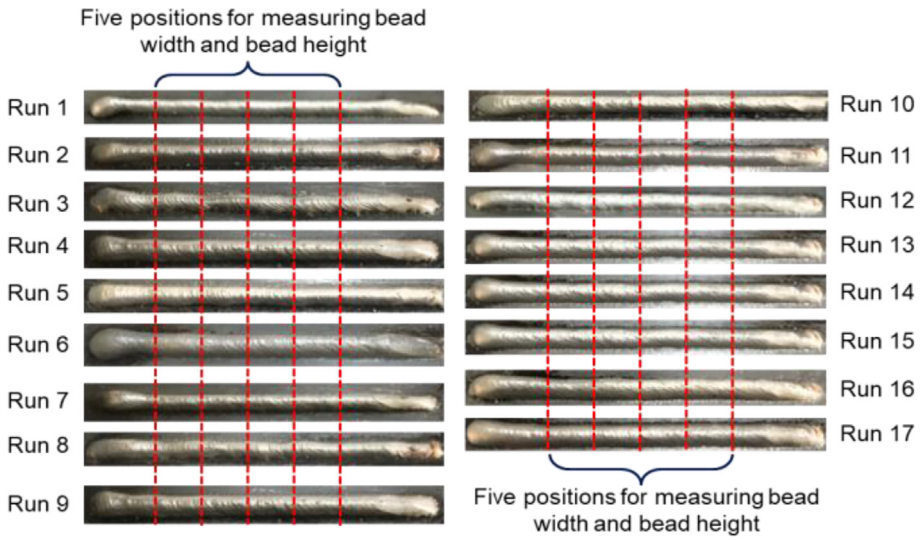


Fig. 1. Images of single layer welds performed according to the experimental design matrix (Table 3).

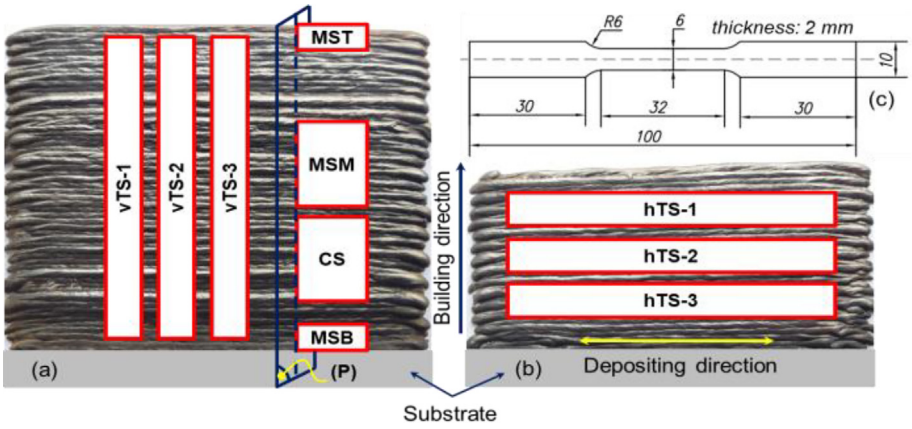


Fig. 2. GMAWAM thin-wall samples: (a) the wall for preparing three microstructure test specimens (MST, MSM, and MSB), a chemical composition test specimen (CS), and three vertical tensile specimens (vTS-1, 2, and 3), (b) the wall for preparing three horizontal tensile specimens (hTS-1, 2, and 3), and (c) tensile specimen dimensions.

Table 4
Chemical composition of the 308L steel wall built by GMAWAM.

Element	C	P	S	Cr	Ni	Mo	Cu	Si	Mn	Al	Fe
GMAWAM 308L steel wall	0.021	0.028	0.019	19.86	9.84	0.0096	0.076	0.48	1.70	–	Balance

Table 5
Results of ANOVA for the regression models.

Factors	Bead width (*BW*)			Bead height (*BH*)		
	Contribution	F-value	p-value	Contribution	F-value	p-value
Model	99.01%	78.06	< 0.0001	99.62%	200.86	< 0.0001
I	3.93%	27.91	0.0011	38.27%	694.76	< 0.0001
U	70.18%	497.94	< 0.0001	10.62%	192.51	< 0.0001
v	18.54%	131.56	< 0.0001	48.21%	875.13	< 0.0001
$I \times U$	0.03%	0.2416	0.6381	0.00%	0.0003	0.9856
$I \times v$	0.07%	0.4736	0.5135	0.19%	3.24	0.1149
$U \times v$	0.42%	2.98	0.1280	0.02%	0.2937	0.6047
I^2	3.89%	25.84	0.0014	0.95%	20.05	0.0029
U^2	1.85%	13.40	0.0081	1.15%	19.74	0.0030
v^2	0.10%	0.6767	0.4379	0.22%	4.01	0.0853
R^2	0.9901			0.9961		
Adj. R^2	0.9774			0.9912		
Pred. R^2	0.8726			0.9706		

regression models of the bead width and the bead height. Their accuracy was evaluated by the determination coefficients, i.e. "R^2", "Adj. R^2", and "*Pred.* R^2". The *p*-value less than 0.05 indicated the significant terms of the developed models, whereas the *p*-value superior to 0.05 indicated the insignificant terms of the developed models.

In the bead width and the bead height models, the terms (I, U, v, I^2 and U^2) with *p*-value less than 0.05 are the significant model terms, and the terms ($I \times U$, $I \times v$, $v \times U$) with *p*-value superior to 0.05 are insignificant. The R^2 values of the bead width and the bead height models are 0.9901 and 0.9961, respectively, indicating a good relationship between the predicted and the experimental values. The "*Pred.* R^2" in the bead width and the bead height models are 0.8726 and 0.9706, respectively, which are in logical agreement with the "*Adj.* R^2" values (0.9774 for the bead width model and 0.9912 for the bead height model). Therefore, the models of the bead width and the bead height are validated in the whole design space, and can be used to predict the optimized processing parameters.

3.2. Effects of processing parameters on the geometry of weld beads

Fig. 3a reveals the main effects of the processing parameters on the bead width (*BW*). It is found that the voltage and the travel speed reveal opposed effects on the bead width. The bead width increases when the voltage increases from 17 V to 23 V, whereas the bead width decreases with an increase in the travel speed from 300 mm/min to 500 mm/min. Moreover, the bead width gradually increases when the welding current increases up to a value around 120 A, subsequently it remains approximately constant. Among these processing parameters, the voltage also has the highest effect on the bead width with a contribution of 70.18%, followed by the travel speed (18.54%) and the current (3.93%), as shown in Table 5. In the case of the bead height (Fig. 3b), the bead height increases when the welding current increases from 100 A to 140 A. In contrast, the bead height decreases when the voltage and the travel speed increase. The travel speed has the highest effect on the bead height with a contribution of 48.21%, followed by the welding current (38.27%) and the voltage (10.26%), Table 5.

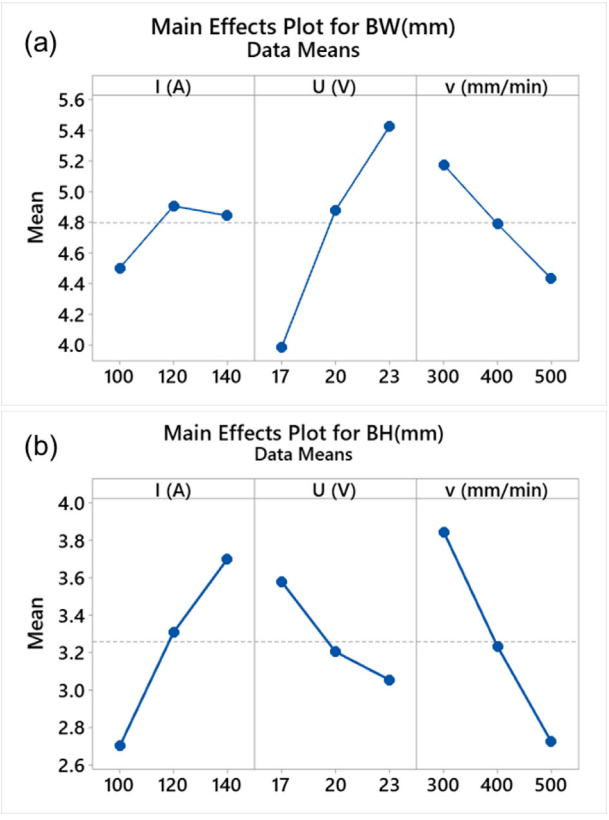


Fig. 3. Main effects of processing parameters on the bead width (a) and on the bead height (b).

Fig. 4a–c exhibits the interaction effects of the processing parameters on the bead width (*BW*). It is revealed that the bead width increases with an increase in the voltage for all values of the welding current and the travel speed (Fig. 4a and c). The bead width always decreases with an increase in the travel speed for all values of the voltage and the welding current (Fig. 4b and c). For all values of the travel speed, the weld bead width only increases when the welding current increases up to a value between 120 A and 130 A, and then it starts decreasing (Fig. 4b). On the other hand, for all values of the voltage, the weld bead width decreases when the welding current increases up to a value between 120 A and 130 A, and then it starts increasing (Fig. 4a). Fig. 4d–f shows the interaction effects of the processing parameters on the bead height (*BH*). The bead height increases when the current increases for all values of U and v. On the other hand, the bead height decreases with an increase in the travel speed for all values of I and U.

3.3. Process parameter optimization

In the GMAWAM process, the bead geometry and the quantity of heat input play very important role. The weld beads with a large width and no major height variation are usually desirable [22]. This ensures the stability of deposits. Besides, the heat input should be minimized to reduce the residual stresses and distortion of the built walls, and to enhance mechanical properties [27]. The heat input was determined by the formula: $Q = 60\eta IU/v$ (in J/mm), where η is the thermal efficiency coefficient, $\eta = 0.8$ for the GMAW [28]. Hence, the problem of optimizing the processing parameters was expressed as follows:

Find [I, U, v] to maximize *BW*, maximize *BH*, and minimize *Q*,

Subject to: $100 \leq I \leq 140$ (A); $17 \leq U \leq 23$ (V); $300 \leq v \leq 500$ (mm/min).

This multi-objective optimization problem was solved by using the optimization module in the Minitab 19 software. The optimal process parameters were calculated as: $I = 122$ (A), $U = 20$ (V) and $v = 368$ (mm/min). Fig. 5 shows three single weld beads generated with these optimized process parameters. They are nearly identical. All weld beads are continuous, smooth and without major defects.

The average value of the bead height and the bead width in the steady region of these weld beads were measured (Table 6). This indicates that the optimal process parameters allow good weld bead shape and size for the GMAWAM process to be achieved. The optimal process parameters were then used to build two single-bead multi-layer thin-walled samples (Fig. 2) for the second stage of this study.

3.4. Part quality

As presented in Figs. 2 and 6, the GMAWAM 308L walls reveal relatively regular height. The first wall (Fig. 2a) shows an average height of 102.38 ± 1.07 mm, whereas the second wall (Fig. 2b) reveals an average height of 58.82 ± 1.25 mm. The first wall was built with 56 deposited layers, thus the average layer thickness is around 1.82 mm. For the second wall, which was built with 30 deposited layers, the average layer thickness is around 1.96 mm. The average layer thickness of the walls is lower than the bead height of single beads (Table 6), because the top of the previously

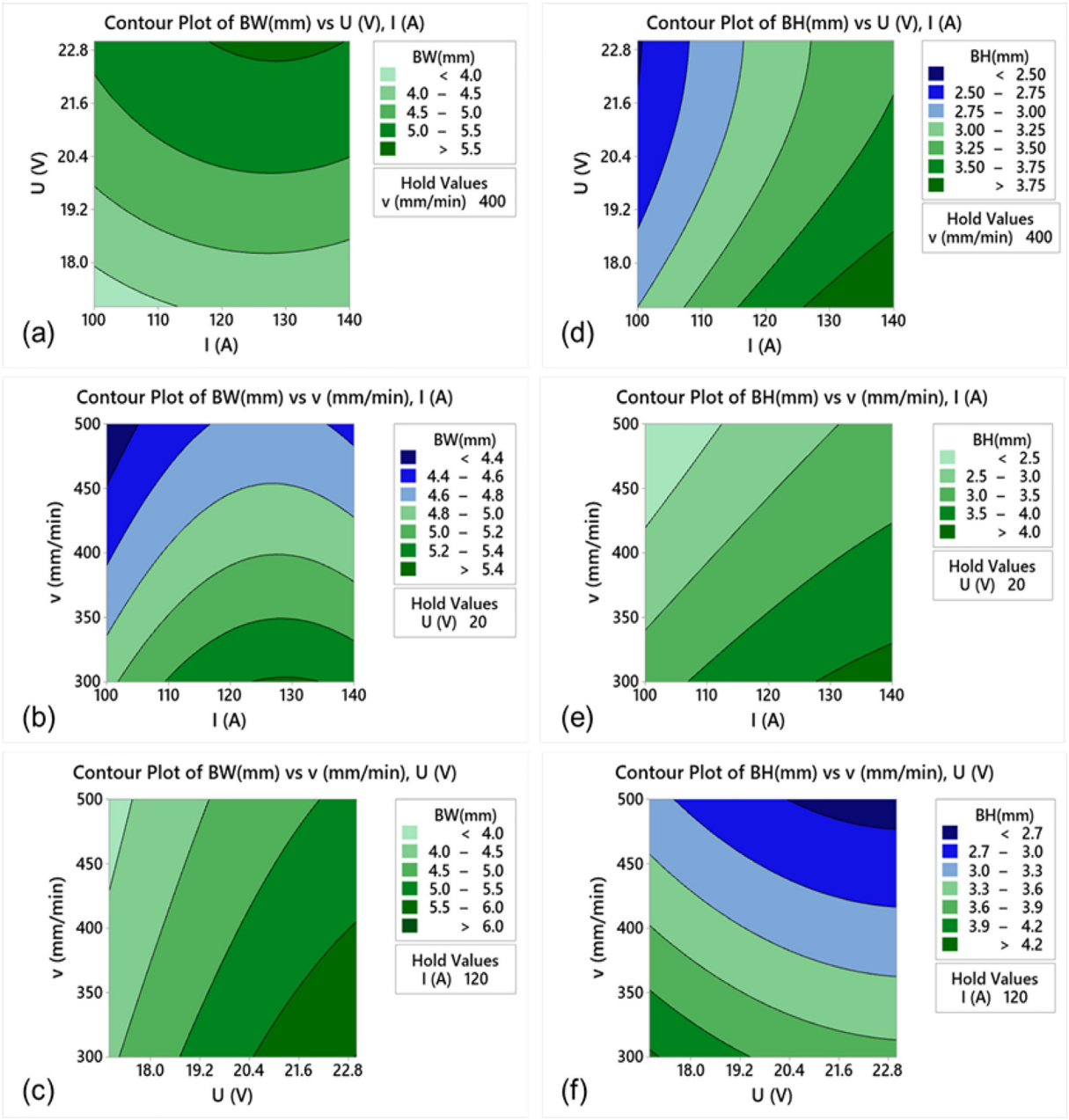


Fig. 4. Interaction effects of processing parameters on the bead width (a–c) and on the bead height (d–f).



Fig. 5. Three single weld beads produced with the optimized processing parameters.

deposited layer was remelted and flattened by the heat source, which forms the subsequent layer during the GMAWAM process. The walls also reveal the wavy morphology on both side surfaces.

Table 6
Dimensions of three single weld beads produced by the optimized process parameters.

Single welding bead	Opt. 1	Opt. 2	Opt. 3
Bead width (mm)	4.98 ± 0.14	4.92 ± 0.11	4.95 ± 0.09
Bead height (mm)	3.34 ± 0.05	3.36 ± 0.02	3.32 ± 0.03

Fig. 7a presents the surface appearance of the second wall (Fig. 2b) obtained by a non-contact 3D digitizer (KONICA MINOLTA RANGE 7) in the effective area of the side surface (Fig. 8). To evaluate the surface roughness, the digitized data (STL file) of the side surface was transformed to the point cloud {pi}, and a plane (Q) that fits the point cloud {pi} was estimated (Fig. 7a). The equation of the plane (Q) is described by Eq. (4):

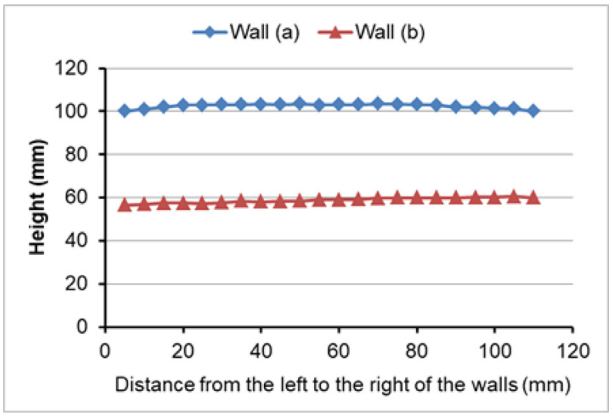


Fig. 6. The height variation of the walls (a) and (b) in Fig. 2 built with the optimized process parameters.

$$(Q): \alpha X + \beta Y + \gamma Z + \delta = 0 \tag{4}$$

where α, β, γ, and δ are the coefficients of the plane and $\alpha^2 + \beta^2 + \gamma^2 > 0$.

Once the plane (Q) was obtained, the surface roughness (SR) of the side surfaces was estimated by Eq. (5):

$$SR = \frac{1}{N_p}\sum_{i=1}^{N_p} d_i = \frac{1}{N_p}\left(\sum_{i=1}^{N_p} \frac{|\alpha X_i + \beta Y_i + \gamma Z_i + \delta|}{\sqrt{\alpha^2 + \beta^2 + \gamma^2}}\right) \tag{5}$$

where N_p is the number of the points in the point cloud $\{\mathbf{p_i}\}$ and d_i is the distance from the point $\boldsymbol{p_i}$ (X_i, Y_i, Z_i) to the fitted plane (Q). The cartography of the distances (d_i) in the computation area was shown in Fig. 7c. The surface roughness of the wall was about 0.23 mm. This value is similar to those reported in the study of [29].

The effective dimensions of the second wall were also measured (Fig. 8): the effective length, EL = 96 mm, the effective height, EH = 56 mm, and the effective width, EW = 3.56 mm. Thereby, the effective volume of the wall material was calculated as: $EV = EL \times EH \times EW = 19138.56$ mm^3. The total volume of the deposited materials for building the wall (TV) is calculated as: $TV = L \times v_w \times (\pi d^2/4)/v$, where L is the total length of the deposits, herein L = 115 mm × 30 layers = 3450 mm, v_w is the wire feed speed, v_w = 2850 mm/min, d is the wire diameter, d = 1 mm, and v is the travel speed, v = 368 mm/min. Thus, TV is approximately equal to 20974 mm^3, and the deposition efficiency, DE = 100% × EV/TV = 91.25%. This demonstrates that the thin walls built with optimal process parameters reveal high deposition efficiency and comparable to those reported in previous studies [30].

Furthermore, X-ray computed tomography (CT) experiments were performed on all vertical and horizontal tensile specimens.

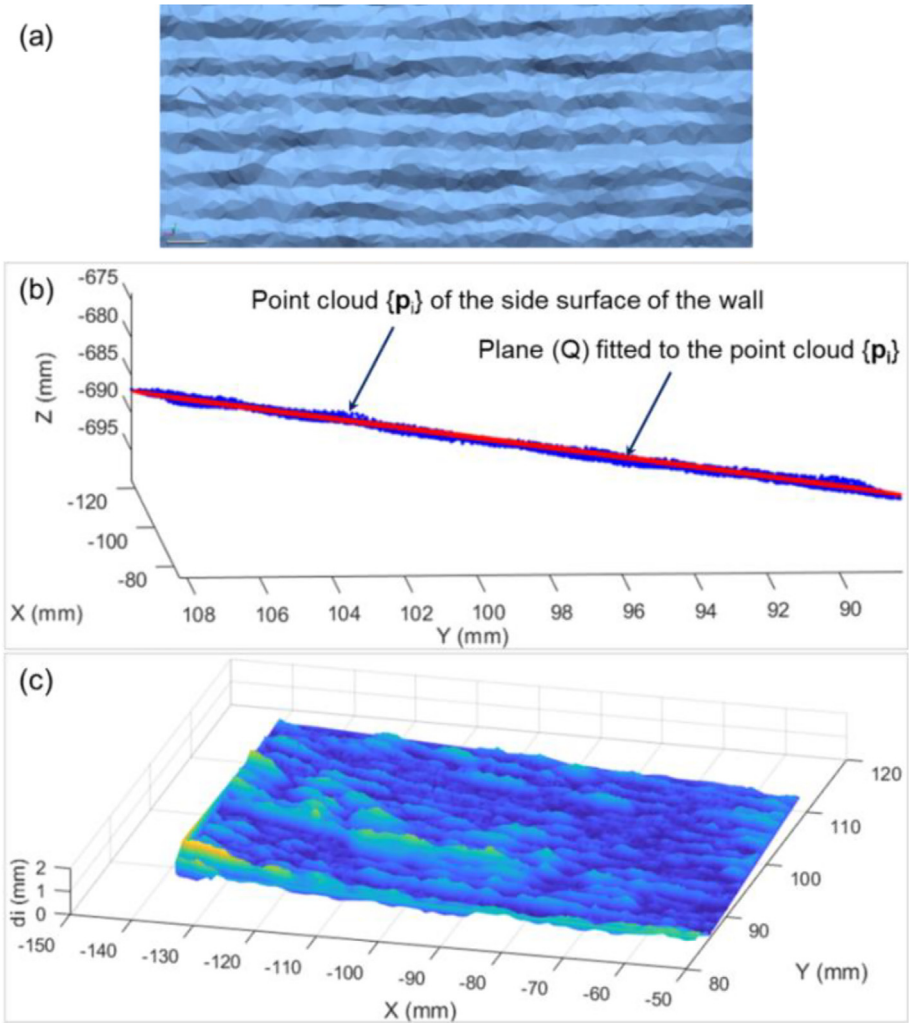


Fig. 7. Surface roughness computation for the side surfaces of GMAWAM walls: (a, b) point cloud $\{\mathbf{p_i}\}$ obtained by laser scanning of the side surfaces and the fitting plane (Q), and (c) cartography of distance d_i from the point $\mathbf{p_i}$ to the plane (Q).

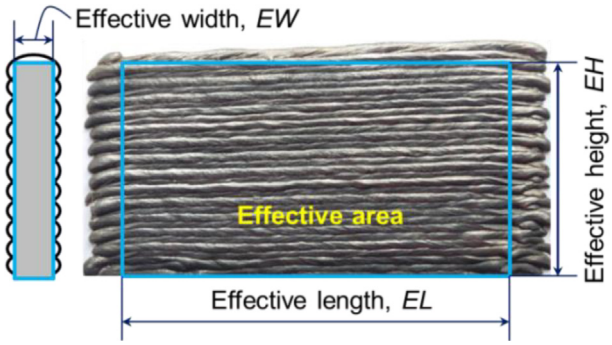


Fig. 8. Schematic diagram illustrating the effective dimensions of the wall.

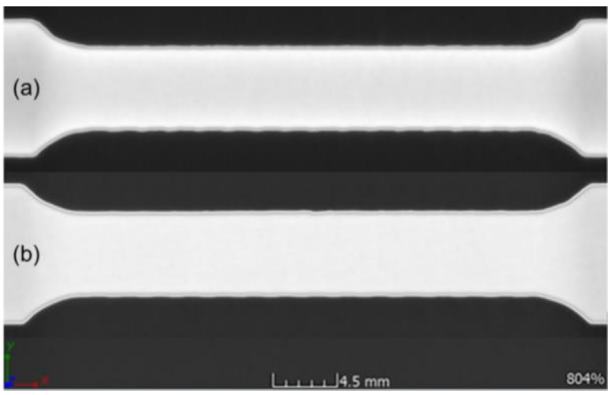


Fig. 9. X-ray CT photograph of a vertical tensile specimen (a) and a horizontal tensile specimen (b).

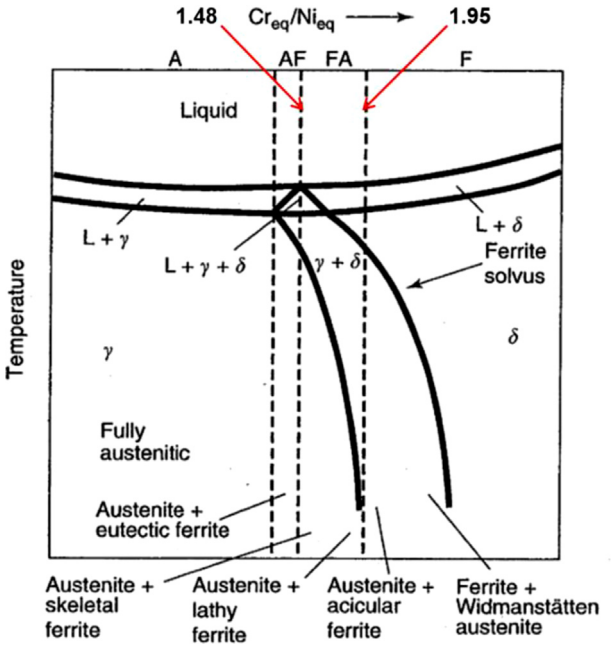


Fig. 10. Relation between the pseudo-binary phase diagram and the solidification mode for austenitic stainless steels [32].

Fig. 9 shows typical X-ray CT images of tensile specimens extracted from the walls in the vertical and horizontal directions. In these images, the tensile specimens appear in white, and macro porosity and cracks normally appear black in images. It is revealed that there are no black zones that are detected in the X-ray CT images. This signifies that all tensile specimens do not contain any major defects, for example macro porosity and cracks. In other words, the wall was successfully built without major defects.

3.5. Microstructures of thin-walled 308L steel components

The microstructure of the GMAWAM 308L walls was observed on the cross-sections of the wall (i.e. the plane (P) shown in Fig. 2a) in three zones: the top zone (specimen MST), the middle zone (specimen MSM) and the bottom zone (specimen MSB). Overall, it is observed that the microstructure of GMAWAM 308L walls dominantly consists of austenite dendrites γ and residual ferrites δ existing within austenitic matrix (Figs. 11–13).

For austenite stainless steels, the solidification mode is identified by the Creq/Nieq ratio, where Creq and Nieq are chromium and nickel equivalent, respectively. The percentages of Ni_{eq} and Cr_{eq} are determined by the formula of Schaeffler [31]: $Ni_{eq} = Ni + 0.5 \times Mn + 30 \times C$ and $Cr_{eq} = Cr + 0.5 \times Nb + 1.5 \times Si + Mo$. Based on the chemical elements of GMAWAM 308L walls obtained from the chemical analysis (Table 4), the percentage of Ni_{eq} and Cr_{eq} are 11.32% and 20.59% respectively; thereby, $Cr_{eq}/Ni_{eq} = 1.82$. As this value falls in range of 1.48–1.95, the solidification of GMAWAM 308L stainless steel occurs according to the Ferrite-Austenite (FA) mode (Fig. 10, [32]): Firstly, the ferrites δ are generated from the liquid phase L, and grow as the primary phase. Subsequently, because of the rapid cooling, the austenite phases γ grow and gradually replace the primary ferrite. The residual ferrites exist in the grain boundaries of austenite with skeletal and lathy morphologies (Figs. 11–13).

As the thermal cycle, the gradient of temperature, and the rate of solidification are different in the top, the middle and the bottom zones of the wall, the microstructure of GMAWAM 308L walls changes in terms of size and morphologies from zone to zone. Fig. 11 shows the microstructure from the center of the cross-section in the bottom zone (specimen MSB). The columnar dendrites which grow in different directions are dominant in this zone, and the grain size is finer than other zones. The reason is that this zone has the highest cooling rate as the heat is transferred to the cold substrate [33]. Moreover, when the first layer is deposited and solidified, the substrate becomes the heterogeneous nucleation and the polycrystalline surface, leading to the growth of dendrites in different directions and orientations [18]. The ferrite phases in this zone are revealed in both lathy and skeletal morphologies (Fig. 11).

The microstructure in the middle zone (specimen MSM) is composed of columnar dendrites (Fig. 12) which grow nearly in the vertical orientation. Moreover, in this zone, the layer $(N-1)$ is partially remelted by the heat which forms the layer N. Hence, it is also observed the heat-affected zone (HAZ) between the layers

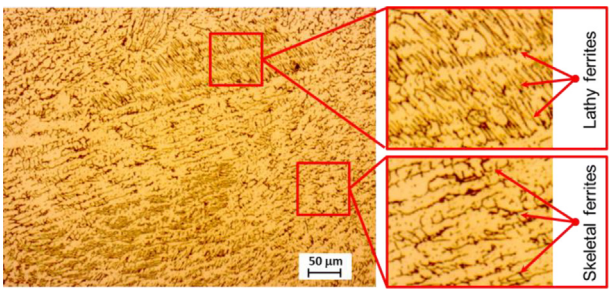


Fig. 11. Microstructures of GMAWAM thin-walled 308L in the bottom zone: columnar dendrites growing in different directions and residual ferrites existing in the grain boundaries with skeletal and lathy morphologies.

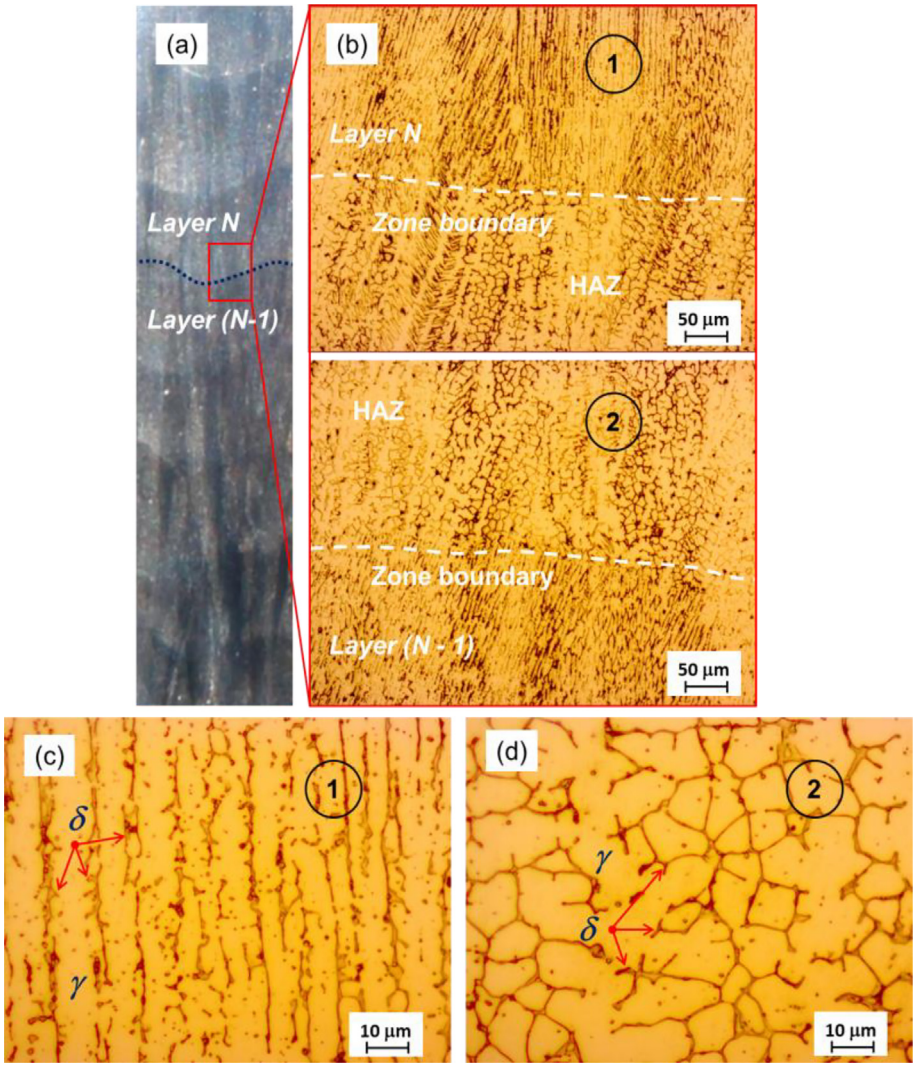


Fig. 12. Microstructures of GMAWAM thin-walled 308L in the middle zone: (a) low magnification photograph, (b) typical microstructure of the middle zone, (c) high magnification of microstructures in the body of a layer, and (d) high magnification of microstructures in the HAZ between the layers ($N-1$) and N.

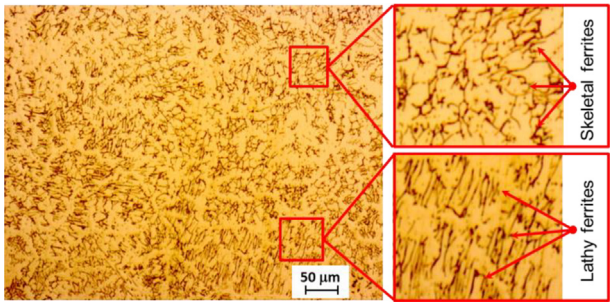


Fig. 13. Microstructures of GMAWAM thin-walled 308L in the top zone: granular grains of austenite and residual ferrites existing in boundaries of austenite matrix with skeletal and lathy morphologies.

($N-1$) and N, the zone ② (Fig. 12b). The HAZ generally consists of equiaxed grains (Fig. 12d) that are coarser than that in the body zone ① of a deposited layer (Fig. 12c). Therefore, a specified layer in the middle zone of GMAWAM 308L walls is characterized by finer columnar grains growing nearly vertically from the bottom of the layer and then transformed into coarse columnar structures with equiaxed grains in the top of the layer (i.e. HAZ), as reported in Ref. [14]. In this zone, the residual ferrite phases mainly exhibit in skeletal and vermicular morphologies. Additionally, the grains in this zone are coarser than those of the bottom zone. This is due to the fact that the gradient of temperature and the rate of cooling in this zone are lower than those in the bottom zone. The layers in the middle zone are deposited on the previously deposited layers that are obviously warmer than the substrate. The heat is also accumulated when the deposited layers increase. As a result, the middle zone features slower cooling rate and longer solidification growth.

In the top zone (specimen MSB), the microstructure from the center of cross-section consists of equiaxed grains of austenite and residual ferrite phases with lathy and skeletal morphologies, which exist in the boundaries of austenite (Fig. 13). In comparison to the microstructure in the middle zone, the grains of the top zone are relatively coarser. Indeed, the top of the molten pool of the last layer contacts directly the natural air, it was solidified into steering dendrites. On the other hand, the bottom of molten pool was solidified into columnar dendrites according to the epitaxial growth. The final solidification zone is the center of the layer. Because

the temperature gradient is suppressed, the center of the top layer is transferred into equiaxed grains [34].

3.6. Mechanical properties

The average microhardness values of the top, the middle, and the bottom zones are 155 ± 1.20, 163 ± 5.36, and 169 ± 5.67 HV0.1, respectively. Additionally, in the middle zone, the HAZ exhibits lower average microhardness than that in the center (body) of a deposited layer. This microhardness variation is in reasonable agreement with the microstructure observed in Section 3.5, and can be explained by the Hall-Petch relationship [35]. According to this relationship, the yield strength and harness of materials will be higher in zones with finer grains in microstructure [14]. Thus, the bottom zone of the wall exhibits the highest average value.

Table 7
Summary of tensile test results.

Specimen	Yield strength (YS, MPa)	Ultimate tensile strength (UTS, MPa)	Elongation (EL, %)
vTS-1	336	537	41
vTS-2	344	531	38
vTS-3	351	528	40
Average	343.67 ± 7.53	531.78 ± 4.52	39.58 ± 1.38
hTS-1	347	553	55
hTS-2	349	558	53
hTS-3	362	548	55
Average	352.69 ± 8.12	552.95 ± 4.96	54.13 ± 1.29

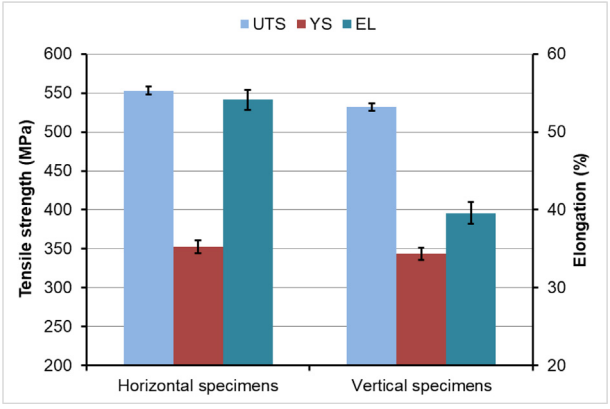


Fig. 14. Comparison of tensile properties (UTS, YS, and elongation) between the vertical and horizontal specimens.

The middle zone reveals the microstructure with grain size relatively finer than the top zone, so this zone is harder than the top zone. This variation trend in terms of microhardness is also similar to that observed in previously published studies [14,36].

Table 7 summarizes the results obtained from the tensile tests for three vertical tensile specimens (vTS-1, vTS-2, vTS-3) and three horizontal tensile specimens (hTS-1, hTS-2, hTS-3). A comparison of tensile properties between the vertical and the horizontal directions is also presented in Fig. 14. In the depositing direction, the UTS and the YS of GMAWAM 308L walls are 552.95 ± 4.96 MPa and 352.69 ± 8.12 MPa, respectively, whereas YS = 343.67 ± 7.53 MPa and UTS = 531.78 ± 4.52 MPa in the building direction. The elongation (EL) of the horizontal specimens is also higher than that of the vertical specimens (54.13 ± 1.29% vs. 39.58 ± 1.38%). This anisotropy phenomenon in tensile properties is due to the inhomogeneity of microstructures [14]. The upward growth of columnar crystals in the building direction is the main reason causing anisotropy of tensile properties between the horizontal and vertical directions. The horizontal tension generally resists grain boundary forces, whereas the vertical tension mainly resists the forces inside the grains and the bonding forces between deposited layers. The results of tensile properties obtained in this study are also similar to those reported by Abioye [17], in which the thin-walled 308LSi parts were produced by a laser metal deposition using wire feeding system. However, the anisotropy trend of tensile properties in this study is contrary to those of 316L thin walls manufactured by WAAM [4,37], in which the authors showed that the tensile strengths in the vertical direction are higher than those in the horizontal direction. Lastly, the tensile strengths of GMAWAM 308L walls fall in the range of values of wrought 308L steel (UTS: 530–650 MPa, YS: 360–480 MPa, and EL: 35–45%). This observation confirms that the internal quality of GMAWAM 308L parts is adequate for industrial applications.

Fig. 15 reveals the morphologies of the fracture surfaces of horizontal tensile specimens (Fig. 15a) and vertical tensile specimens (Fig. 15b). It is revealed that there are many dimples with homogenous distribution on the fracture surfaces of the horizontal and vertical tensile specimens. This demonstrates that the GMAWAM 308L features a good toughness [37]. Moreover, the dimples' depth and size observed in the facture surface of the horizontal tensile specimens (Fig. 15a) are obviously more refined than those of the vertical specimens (Fig. 15b). This explains why the toughness in the depositing direction is higher than that in the building direction.

4. Conclusions

In the current research, the manufacture of 308L walls by GMAWAM has been investigated. The main results of the study are summarized, as follows:

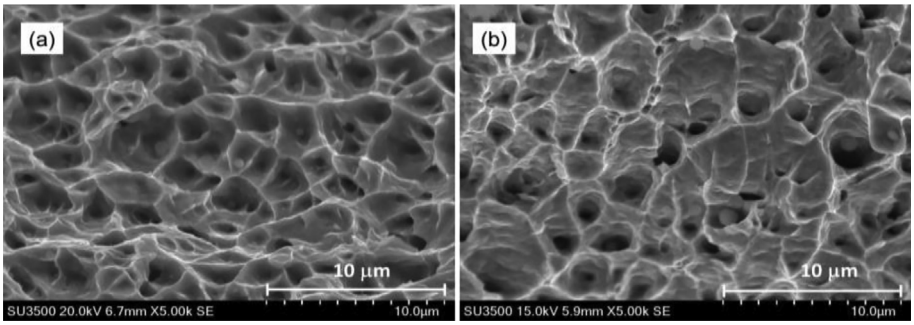


Fig. 15. Fracture surface morphologies scanned by the SEM machine of a horizontal tensile specimen (a) and a vertical tensile specimen (b).

- The main processing parameters of the GMAW process, including the welding current, the voltage, and the travel speed significantly influence the size and shape of single weld beads. The voltage and the travel speed have major effects on the bead width, while the welding current and the travel speed have significant effects on the bead height.
- The optimized processing parameters (I = 122 A, U = 20 V and v = 368 mm/min) were estimated by using the response surface methodology (RSM). These optimal parameters enable us to produce smooth and regular single weld beads with desired bead width and bead height, while minimizing the heat input. This ensures the stability and the quality of 308L walls built by GMAWAM.
- The GMAWAM 308L walls were built successfully without any major defect such as cracks. Strong bonding was also observed between deposited layers. The height of the walls is relatively regular and the material deposition efficiency can reach 91%.
- The typical microstructure of GMAWAM 308L walls consists of two phases: the columnar austenite dendrites and residual ferrite existing within the austenitic matrix. In a specific deposited layer, the columnar dendrites epitaxially grow from the remelted zone of the previous layer along the build direction, and then transformed into coarse columnar structures with equiaxed grains at the top of the layer.
- The average microhardness of GMAWAM 308L walls is about 163 ± 5.36 HV0.1.
- The tensile strengths of GMAWAM 308L walls show an anisotropic characteristic: YS = 343.67 ± 7.53 MPa, UTS = 531.78 ± 4. 52 MPa, and EL = 39.58 ± 1.38% in the building direction, while YS = 352.69 ± 8.12 MPa, UTS = 552.95 ± 4.96 MPa, and EL = 54. 13 ± 1.29% in the depositing direction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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